

Disjoint paths in a connected star friends-and-strangers graph

Theorem

Let X be a finite simple graph on $[n]$, let Star_n be the star on $[n]$ with center n , and put

$$G = \text{FS}(X, \text{Star}_n).$$

Its vertices are the bijections $\tau : [n] \rightarrow [n]$, with $\tau(i')$ giving the person at position i' . Two vertices differ by a friendly swap when they exchange the people at adjacent positions of X , and one of the two exchanged people is n . Thus we use exactly the position-to-person convention of Krishnan and Li.

Assume that G is connected. If $\sigma, \rho \in V(G)$ are distinct, then the maximum number of pairwise internally vertex-disjoint σ - ρ paths is

$$\min\{\deg_G(\sigma), \deg_G(\rho)\}.$$

Here, as in the source paper, paths may meet at their two common endpoints but nowhere else. If σ and ρ are adjacent, the edge $\sigma\rho$ itself counts as one path.

A connectivity theorem for Bi-Cayley graphs

We use the following theorem of Liang and Meng. Let H be a finite group and $S \subseteq H$. The Bi-Cayley graph $\text{BC}(H, S)$ has two copies of H as its bipartition classes, with $(g, 0)$ adjacent to $(sg, 1)$ for $g \in H$ and $s \in S$. If $\text{BC}(H, S)$ is connected, then its vertex connectivity is its degree $|S|$ [Liang–Meng, Theorem 3.7].

We will use the equivalent formulation that if a finite connected bipartite graph has a group of automorphisms acting regularly on each bipartition class, then its vertex connectivity equals its degree. Indeed, after choosing one vertex in each class, the two regular actions identify both classes with the group, and invariance under the action gives precisely a Bi-Cayley graph. If these identifications give edges from $(g, 0)$ to $(gs, 1)$ instead, replacing both group coordinates by their inverses gives the convention above, with connection set S^{-1} .

Proof strategy

We first fix a type h and prove the result for two permutations of that same type. Delete the whole type fibre T_h . Each component left behind represents one possible class of excursions during which the center n leaves position h and later returns. Contracting those components gives a bipartite incidence graph J_h .

The graph J_h is generally only biregular, so the Bi-Cayley connectivity theorem does not apply to it directly. We use the symmetry among the ordinary people to show that every component has the same multiplicity q , and then replace each component-node by q actual representatives. The resulting regular graph K_h is a connected Bi-Cayley graph. Its connectivity transfers back to J_h , and disjoint paths in J_h lift through the original components to disjoint paths in G . This settles equal-type endpoints.

Finally, a simple path in X has $(n-1)!$ mutually disjoint lifts in G , one from each permutation of its initial type. These lifts, together with the equal-type result, join arbitrary endpoint types. When the endpoints are adjacent, we reserve their edge and apply the same argument after deleting that edge.

Proof

The cases $n = 1$ and $n = 2$ are immediate: the first is vacuous, and in the second a connected G is a single edge. Hence assume $n \geq 3$.

The source paper calls

$$\text{type}(\tau) = \tau^{-1}(n) \in V(X)$$

the type of a permutation $\tau \in V(G)$. Thus the type is the position occupied by the center n of Star_n , or equivalently the empty position in the sliding-puzzle interpretation. A friendly swap moves n to a neighboring position in X . Consequently

$$\deg_G(\tau) = \deg_X(\text{type}(\tau)). \quad (1)$$

For $x \in V(X)$, write

$$T_x = \{\tau \in V(G) : \text{type}(\tau) = x\}.$$

Each T_x has $(n-1)!$ elements and is an independent set in G .

1. Deleting any one position from X

First, X is connected. Indeed, under any walk in G , the sequence of types is a walk in X ; and every vertex of X occurs as the type of some permutation. If X had two components, permutations whose types lie in different components could not be joined in G .

We next record the stronger consequence that, for every $h \in V(X)$, the graph $X - h$ is connected.

Suppose instead that $X - h$ has at least two components. Consider a walk in G whose first and last permutations lie in T_h . Split the walk into excursions between consecutive visits to T_h . During one such excursion, the type leaves h , stays in a single component C of $X - h$, and eventually returns to h . The first move carries one ordinary person from C to h , and the last move carries that same person back into C . All intervening moves take place inside C . Therefore the set of ordinary people occupying the positions of C is unchanged by the excursion. The corresponding set is also unchanged for every other component of $X - h$.

It follows that along every walk between two permutations in T_h , the set of people in each component of $X - h$ is invariant. But two permutations in T_h can be chosen by exchanging people in two different components of $X - h$. They could not be joined in G , contrary to the connectedness of G . Thus $X - h$ is connected.

In particular, every vertex of X has degree at least two. Connectedness of X rules out degree zero. If x were a leaf with neighbor y , then x would be isolated in $X - y$, which is impossible when $n \geq 3$.

2. Delete one type fibre and form an incidence graph

Fix $h \in V(X)$ and set

$$D = \deg_X(h).$$

We will first prove the theorem for two permutations of type h . The reason for deleting the entire fibre T_h is that, once a walk leaves type h , the ordinary person left at position h cannot change until the walk returns to type h . The components of $G - T_h$ therefore retain exactly the information needed to describe excursions away from h .

Let $\mathcal{B}_h$ be the set of connected components of $G - T_h$. Define a bipartite incidence graph J_h with bipartition

$$T_h \sqcup \mathcal{B}_h.$$

A permutation $\tau \in T_h$ is adjacent in J_h to $B \in \mathcal{B}_h$ when τ has a neighbor in B in the original graph G . This is a simple graph: adjacency records only whether such an edge exists. Equivalently, contract every component of $G - T_h$ to one vertex and retain the vertices of T_h . This gives J_h , so J_h is connected because G is connected.

Fix $B \in \mathcal{B}_h$. Every permutation $\beta \in B$ has the same ordinary person at position h . Indeed, an edge of $G - T_h$ cannot swap at h , since such a swap would have one endpoint of type h . Denote this fixed person by

$$\ell(B) = \beta(h) \in [n-1].$$

The component B and the person $\ell(B)$ are different objects: several components could a priori have the same fixed person. We use $\ell(B)$ only to distinguish the possible attachments of B to T_h .

The component B contains permutations of every type $x \neq h$. To see this, start from any $\beta \in B$ and take a path in the connected graph $X - h$ from $\text{type}(\beta)$ to x . Moving n along this path gives a path in $G - T_h$, so its endpoint remains in B and has type x .

More is true: the number of permutations of each type $x \neq h$ in B is independent of x . If P is a fixed path from x to y in $X - h$, moving n along P maps the type- x permutations in B bijectively to the type- y permutations in B ; moving along the reverse path is the inverse map. Write this common positive number as

$$q_B = |B \cap T_x| \quad (x \neq h). \quad (2)$$

We now compute the two degrees in J_h . If $\tau \in T_h$, moving n from h to each of its D neighbors in X produces D neighbors of τ in $G - T_h$. The D moves leave D different people at position h , so their endpoints belong to different components B . Hence

$$\deg_{J_h}(\tau) = D. \quad (3)$$

On the other side, let $B \in \mathcal{B}_h$. For every neighbor x of h , there are q_B permutations in $B \cap T_x$, and each is adjacent in G to a permutation in T_h . These Dq_B permutations give Dq_B distinct neighbors of B in J_h . The point requiring verification is that two of them cannot lead to the same $\tau \in T_h$. If τ is adjacent to B , then the move from τ into B must swap h with the unique position x at which $\tau(x) = \ell(B)$. Thus x , and then the resulting neighbor of τ in B , is uniquely determined. Therefore

$$\deg_{J_h}(B) = Dq_B. \quad (4)$$

In particular, J_h has minimum degree D , but it need not be regular because q_B has not yet been shown to be constant and may exceed one.

3. Symmetry and regularization

Let

$$\Gamma = \mathfrak{S}_{[n-1]}$$

be the group permuting the ordinary people and fixing n . It acts on $V(G)$ by $\tau \mapsto \gamma \circ \tau$. This action preserves friendly swaps, hence gives automorphisms of G . It is transitive, in fact regular, on every type fibre T_x .

The same action is transitive on $\mathcal{B}_h$. Choose one position $p_0 \neq h$. By the preceding paragraph, every $B \in \mathcal{B}_h$ contains a permutation of type p_0 . Given $B_1, B_2 \in \mathcal{B}_h$, choose $\beta_i \in B_i \cap T_{p_0}$. There is a unique $\gamma \in \Gamma$ such that $\gamma \circ \beta_1 = \beta_2$. Since γ preserves T_h , it maps

components of $G - T_h$ to components and therefore maps B_1 onto B_2 . It follows from (2) that all q_B have a common value, say q .

The incidence graph J_h is now (D, Dq) -biregular. We cannot apply the regular Bi-Cayley theorem directly when $q > 1$. We therefore replace every component-node B by its q actual representatives in $B \cap T_{p_0}$.

Define an auxiliary bipartite graph K_h with bipartition

$$T_h \sqcup T_{p_0}.$$

For $\tau \in T_h$ and $\beta \in T_{p_0}$, where β belongs to the component $B \in \mathcal{B}_h$, declare $\tau\beta$ to be an edge of K_h exactly when τB is an edge of J_h . Thus the q vertices of $B \cap T_{p_0}$ are q copies of the component-node B , all with the same neighborhood. The graph K_h is auxiliary; its edges need not be friendly swaps in G .

By (3) and (4), every vertex of K_h has degree Dq . It is also connected. Map T_h identically to T_h , and map each $\beta \in B \cap T_{p_0}$ to the node B of J_h . A path in J_h lifts to a path in K_h by choosing, at each component-node B on the path, any representative in $B \cap T_{p_0}$. If an endpoint of the lifted path is a prescribed representative, choose that representative at the corresponding endpoint. Two representatives over the same B are joined through any common T_h -neighbor. It follows that any two vertices of K_h can be joined.

The action of Γ is regular on both T_h and T_{p_0} , and the definition of K_h is invariant under this action. Hence K_h is a connected Bi-Cayley graph over Γ . The Liang–Meng theorem says that the vertex connectivity of K_h is

$$Dq. \tag{5}$$

We transfer this connectivity back to the biregular graph J_h . Suppose that Z is a vertex cut of J_h , and write

$$a = |Z \cap T_h|, \quad b = |Z \cap \mathcal{B}_h|.$$

In K_h , delete the same a vertices of T_h , and for every $B \in Z \cap \mathcal{B}_h$, delete all q vertices of $B \cap T_{p_0}$. Call the resulting set $\hat{Z}$. Then

$$|\hat{Z}| = a + qb \leq q(a + b) = q|Z|. \tag{6}$$

Moreover, a path in $K_h - \hat{Z}$ would project to a walk in $J_h - Z$. Thus vertices lying over different components of $J_h - Z$ cannot be joined in $K_h - \hat{Z}$, so $\hat{Z}$ is a vertex cut of K_h .

If $|Z| < D$, (6) would give $|\hat{Z}| < Dq$, contradicting (5). Therefore every vertex cut of J_h has size at least D . For the reverse bound, delete the D neighbors of any $\tau \in T_h$. This isolates τ , while another vertex of T_h remains because $|T_h| = (n-1)! \geq 2$. Hence the vertex connectivity of J_h is exactly

$$D. \tag{7}$$

4. Permutations of the same type

Let $\sigma, \rho \in T_h$ be distinct. By (7) and Menger's theorem, J_h contains D pairwise disjoint σ - ρ paths. We lift each one to G . Whenever a path passes through a component-node B , its two incident edges specify a neighbor of the preceding T_h -vertex in B and a neighbor of the following T_h -vertex in B . Join these two attachment permutations by a path inside the connected component B , and include the two friendly-swap edges at the ends.

Different paths in J_h have no common internal T_h -vertex and no common component-node B . Their lifts therefore have no common internal vertex in G : different components of $G - T_h$ are disjoint, and none contains a vertex of T_h . Erasing any cycles within an individual lifted walk gives D pairwise disjoint σ - ρ paths in G . By (1), both endpoints have degree D , so no larger family is possible. We have proved

$$\max\{\text{number of disjoint } \sigma\text{-}\rho \text{ paths}\} = \deg_X(h) \tag{8}$$

whenever σ and ρ have the same type h .

5. Arbitrary endpoint types

Now let $\sigma, \rho \in V(G)$ be arbitrary distinct vertices and put

$$\begin{aligned} u &= \text{type}(\sigma), & v &= \text{type}(\rho), \\ d &= \min\{\deg_G(\sigma), \deg_G(\rho)\} \\ &= \min\{\deg_X(u), \deg_X(v)\}. \end{aligned} \tag{9}$$

The case $u = v$ is (8), so assume $u \neq v$.

Choose a simple path

$$P : u = x_0, x_1, \dots, x_r = v$$

in X . Starting from any permutation in T_u , successively move person n along P . This gives a path in G ending in T_v . As the starting permutation ranges over the $(n-1)!$ elements of T_u , these lifted paths are pairwise vertex-disjoint. Indeed, at each fixed level i , moving along the first i edges of P is a bijection from T_u to T_{x_i} , while different levels have different types because P is simple. We therefore have $(n-1)!$ disjoint paths joining the two full fibres T_u and T_v .

First suppose that σ and ρ are not adjacent. Let $Z \subseteq V(G) \setminus \{\sigma, \rho\}$ with $|Z| < d$. By (8), any two surviving permutations in T_u are joined in $G - Z$: among the $\deg_X(u)$ disjoint paths between them, fewer than $d \leq \deg_X(u)$ can meet Z . Hence all of $T_u - Z$ lies in one component of $G - Z$. The same holds for $T_v - Z$.

Since the lifted copies of P are vertex-disjoint and $|Z| \leq d-1 \leq n-2 < (n-1)!$, at least one of them avoids Z . It joins the component containing $T_u - Z$ to the component containing $T_v - Z$, and consequently joins σ to ρ . Thus no set of fewer than d vertices separates σ from ρ . Menger's theorem gives d disjoint paths.

It remains to handle adjacent σ and ρ . Let $e = \sigma\rho$, and reserve e as one of the desired paths. We show that $G - e$ contains $d-1$ further disjoint σ - ρ paths.

Let $Z \subseteq V(G) \setminus \{\sigma, \rho\}$ with $|Z| \leq d-2$. We first check that the vertices of $T_u - Z$ lie in one connected component of the ambient graph $G - Z - e$. Between any two of its vertices, (8) supplies $\deg_X(u)$ disjoint paths in G . At most $|Z|$ of these meet Z , and at most one uses e . For the last assertion, note that a path between two type- u permutations which uses e contains ρ , whose type is v , as an internal vertex; two paths in a disjoint family cannot both do so. Hence at least

$$\deg_X(u) - |Z| - 1 \geq \deg_X(u) - (d-2) - 1 \geq 1$$

paths avoid both Z and e . Therefore all vertices of $T_u - Z$ lie in one connected component of the ambient graph $G - Z - e$. The same argument, with σ as the forced internal vertex of a type- v path using e , places all vertices of $T_v - Z$ in one connected component there as well.

Among the $(n-1)!$ lifted copies of P , at most $|Z|$ meet Z , and at most one uses e , because the lifts are vertex-disjoint. There is still a surviving lift. Indeed, if $n = 3$, then $d = 2$, $Z = \emptyset$, and $2! = (n-1)! > 1$. If $n \geq 4$, then

$$|Z| + 1 \leq d - 1 \leq n - 2 < (n-1)!.$$

This surviving lift joins $T_u - Z$ to $T_v - Z$ in $G - Z - e$, so it joins σ to ρ . We have proved that no set of at most $d-2$ vertices separates σ and ρ in $G - e$. By Menger's theorem, $G - e$ contains $d-1$ disjoint σ - ρ paths. Adding the reserved edge e gives d disjoint paths in G .

Finally, d is also an upper bound. If the endpoints are nonadjacent, disjoint paths must leave either endpoint through distinct neighbors. If they are adjacent, at most one path is the edge $\sigma\rho$, and every other path must leave each endpoint through a different remaining neighbor. Thus no family has more than $\min\{\deg_G(\sigma), \deg_G(\rho)\} = d$ paths. This proves the theorem.

References

N. Krishnan and R. Li, *Vertex Connectivity of Friends-and-strangers Graphs*, Electronic Journal of Combinatorics 33(3) (2026), P3.41; [arXiv:2410.21334v3](#), Conjecture 7.1. The original v1 labels the same conjecture 8.1.

X. Liang and J. Meng, “Connectivity of Bi-Cayley Graphs,” *Ars Combinatoria* **88** (2008), 27–32, Theorem 3.7. [Original article](#).